

Adaptive Maximum-Load Assortments: Comparison Bounds and Truncation

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Abstract

We study adaptive maximum-load assortment optimization under a common multinomial logit model. A capped-choice comparison shows that one iid categorical maximum stochastically dominates the terminal maximum under every adaptive policy. A simple fixed assortment therefore obtains at least half the adaptive optimum, improving the general upper bound on the adaptivity gap from four to two. A two-customer equal-weight construction also raises the lower bound from $4/3$ to a constant exceeding 1.33462 . We construct an accuracy-dependent family of fixed assortments giving a dimension-free large-value certificate; the adaptivity gap tends uniformly to one as the optimal expected load grows. Classical self-bounding concentration applied to the iid reference gives a stopping level of order $U + \log(1/\varepsilon)$ when the adaptive value is at most U . Combined with weight rounding and histogram dynamic programming, this yields a $(1 - \varepsilon)$ approximation in $n^{O_\varepsilon(\log n)}$ time. A bounded-value example rules out retaining only an accuracy-dependent number of previously purchased products in every offered menu. Supplementary results give continuation-state pruning certificates and an alternative proof of static prefix optimality.

1 Introduction and model

There are $n \geq 1$ products with fixed positive weights $v_1, \dots, v_n$, outside weight one, and $T \geq 1$ customers. Conditional on the observed past, a policy selects any assortment $A \subseteq [n]$. A fresh customer chooses $i \in A$ with probability $v_i/(1 + v(A))$, where $v(A) = \sum_{i \in A} v_i$, and chooses outside with the remaining probability. Let L_i be the final inside count and $M_T = \max_i L_i$. The outside count is excluded. Write $F_T(A)$ for the expected maximum under constant menu A , $S = \max_A F_T(A)$ for the static optimum, and $D = D_T = \sup_\pi \mathbb{E} M_T^\pi$ for the adaptive optimum from zero loads. Empty menus and policy randomization are permitted; finite deterministic Bellman policies attain the optimal value. We impose no capacity or inventory constraints and use one common known MNL population.

El Housni, Ibn Brahim, and Segev [1] introduced this problem and gave static approximation algorithms, a general adaptivity-gap upper bound of four, an equal-weight upper bound of two, a lower bound of $4/3$, and a dynamic QPTAS with time $n^{O_\varepsilon(\log^3 n)}$. Their framework supplies the MNL action oracle, fixed-menu evaluation, weight rounding, policy lifting, and histogram dynamic programming used below. We strengthen the comparison and value-reduction ingredients.

Causal majorization couplings are established tools in adaptive occupancy analysis. Azar, Broder, Karlin, and Upfal [3, Lemma 3.4 and Theorem 3.5] compare sampled-candidate online allocations, and Wieder [4, Lemma 2.2] uses rank couplings for heterogeneous bins. Our comparison treats

labeled coordinate probability caps and a maximum-load objective. We claim the joint capped comparison and its MNL consequences, not causal occupancy coupling as a general new technique. Kaplan, Sapir, and Stemmer [5, Lemma 3.3] also compare an adaptive bin-deletion process with nonadaptive iid occupancy; their reference uses a coverage stopping event and redistribution of existing balls. The fixed-horizon capped-addition theorem here has a different feasible control model.

Zhao and Fu [6, Theorem 4.1 and Remark 4.1] prove static top- k optimality and exact polynomial-time optimization, also under an upper cardinality constraint $|A| \leq C$. Their Proposition 4.1 establishes quasiconvexity in one added product's weight, using log-concavity and log-supermodularity of the iid maximum-load distribution, head-start values, and a derivative hierarchy. Their global argument then deletes or upgrades an item. Appendix C reaches the same exchange property through a coefficientwise weak-leader odds inequality. We do not claim priority for static prefix optimality, the exchange consequence, or exact static optimization. The adaptive results below use neither static proof; Section 6 of [6] leaves the adaptive setting as a direction for further study.

Sections 2–5 establish the adaptive comparison, large-value bound, truncation, and approximation scheme. Section 6 concerns a restriction on offered products, not a memory or computational lower bound. Appendix A gives lossless and quantitative continuation-state reductions. The adaptive PTAS and exact adaptivity-gap constant remain unresolved here. Structural and probabilistic statements hold for real positive weights. Algorithmic bit-complexity statements use rational weights and include the numerical horizon T unless an easy branch avoids unrolling it.

2 Capped-choice comparison and the adaptivity gap

Sort $v_1 \geq \dots \geq v_n$. If $v([n]) < 1$, let $A_0 = [n]$. Otherwise let $A_0 = [k]$ be the shortest prefix with $v(A_0) \geq 1$. Put $W = v(A_0)$.

Theorem 2.1. *For every positive weight vector and finite horizon,*

$$F_T(A_0) \geq \beta D \geq \frac{1}{2}D, \quad \beta = \begin{cases} (1+W)^{-1}, & v([n]) < 1, \\ W/(1+W), & v([n]) \geq 1. \end{cases}$$

Consequently $D/S \leq 2$. The same constant menu is valid for every T and is selected using sorting and partial sums only.

The prefix is the one used in the factor-four proof of [1]. What changes is its analysis. It is not necessarily the best static prefix: three unit weights and two customers give value 1 for $A_0 = \{1\}$, whereas the full menu has value $9/8$.

2.1 A capped-choice relaxation

For caps $a = (a_1, \dots, a_n) \in [0, 1]^n$, let

$$\mathcal{K}(a) = \{q \in \mathbb{R}_+^n : q_i \leq a_i, \sum_i q_i \leq 1\}.$$

At each step a controller selects $q \in \mathcal{K}(a)$, outcome i increments load i , and residual probability $1 - \sum_i q_i$ changes no load. This is an *upper-bound model*, not a claim that every capped vector is MNL feasible. For integer initial loads $\ell \in \mathbb{Z}_{\geq 0}^n$, let $J_r^\varphi(\ell)$ be its optimal expectation of $\varphi(\max_i L_i)$, where φ is any nondecreasing function and r steps remain. Only finitely many terminal loads arise, so no global boundedness assumption on φ is needed. Its one-state action problem is a linear program over the compact polytope $\mathcal{K}(a)$.

Lemma 2.2 (Load-cap comparison). *Suppose $a_i \geq a_j$. Let x, y agree outside i, j , have the same pair total, and satisfy $x_i \geq \max\{y_i, y_j\}$. Then $J_r^\varphi(x) \geq J_r^\varphi(y)$ for every r . In particular, if $\ell_i \geq \ell_j$,*

$$J_r^\varphi(\ell + e_i) \geq J_r^\varphi(\ell + e_j).$$

Proof. Fix any policy in the y -system. Couple a feasible x -policy so that states are either equal or retain the displayed concentration relation; the maximum in x is then at least the maximum in y at every time. Once states coincide, use identical future decisions and outcomes. If states differ and $x_i > \max(y_i, y_j)$, use identical outcomes; integrality preserves the relation. The only other case is the swapped boundary $x_i = y_j > y_i = x_j$.

Let q be the y -policy's next capped law. Leave all other probabilities, including the outside probability, unchanged and set

$$p_j = \min\{q_i, a_j\}, \quad p_i = q_i + q_j - p_j.$$

If $q_i \leq a_j$, then $p_j = q_i \leq a_j$ and $p_i = q_j \leq a_j \leq a_i$. Otherwise $p_j = a_j$ and $0 \leq p_i = q_i + q_j - a_j \leq q_i \leq a_i$. Thus $p \in \mathcal{K}(a)$ and the pair probability is preserved.

Couple outcome $y : j$ to $x : i$ with mass q_j ; couple $y : i$ to $x : j$ with mass p_j , and to $x : i$ with mass $q_i - p_j$. All other outcomes agree. Each positive-probability transition retains the relation or makes states equal. This specifies a causal finite coupling. It can be implemented by retaining the virtual y history internally and, after the actual x outcome, sampling its conditional virtual counterpart. No future information is used. Taking terminal expectations and then optimizing proves the first claim. For the second, take $x = \ell + e_i$ and $y = \ell + e_j$. $\square$

Theorem 2.3 (Saturated-group comparison). *Let $H = [k]$, and suppose*

$$a_i = p_i > 0 \ (i \in H), \quad \sum_{i \in H} p_i = 1, \quad a_j \leq \min_{i \in H} p_i \ (j \notin H).$$

At any initial state satisfying $\min_{i \in H} \ell_i \geq \max_{j \notin H} \ell_j$, the constant law p on H with no outside outcome is optimal in the capped process for every nondecreasing terminal function φ . Therefore

$$J_r^\varphi(\ell) = \mathbb{E}\varphi\left(\max_{i \in H}(\ell_i + N_i)\right), \quad (N_i)_{i \in H} \sim \text{Mult}(r, p).$$

The condition is automatic at zero loads; an empty complement causes no restriction.

Proof. At such a state write $d_i = J_{r-1}^\varphi(\ell + e_i) - J_{r-1}^\varphi(\ell)$. Monotonicity of the terminal function, replaying the same history policy, gives $d_i \geq 0$. Lemma 2.2 implies $d_i \geq d_j$ for $i \in H, j \notin H$. Put $b = \min_{i \in H} d_i$. For any feasible next law q ,

$$\begin{aligned} \sum_{i \in H} p_i d_i - \sum_j q_j d_j &= \sum_{i \in H} (p_i - q_i) d_i - \sum_{j \notin H} q_j d_j \\ &\geq b \left(\sum_{i \in H} (p_i - q_i) - \sum_{j \notin H} q_j \right) \\ &= b(1 - \sum_j q_j) \geq 0. \end{aligned}$$

Here $p_i - q_i \geq 0$ follows from the actual head caps; replacing this condition by an aggregate mass bound would not suffice. Hence p is an optimal action. Using it preserves the load condition, because only head coordinates increase. Backward induction gives constant p at every future stage, hence the multinomial formula. The differences d_i were those of the full capped problem, not of a pre-restricted head-only problem, so the argument is not circular. $\square$

Taking $\varphi(z) = \mathbf{1}\{z \geq h\}$ for each integer h yields terminal stochastic domination of *every* capped policy by this fixed multinomial maximum. We do not assert coordinatewise domination of every original product by a fixed head label.

2.2 Jointly bounding the original universe

For any MNL policy, conditioned on its entire past and selected menu,

$$q_i \leq \alpha_i := \frac{v_i}{1 + v_i}.$$

This is a per-round cap only; the adaptive final product counts are not assumed to be independent binomials.

Proof of Theorem 2.1. First suppose $v([n]) \geq 1$. By minimality of k , $W = \sum_{i=1}^k v_i < 1 + v_k$. Define $p_i = v_i/W$ for $i \leq k$. Then

$$p_i > \frac{v_i}{1 + v_k} \geq \alpha_i \quad (i \leq k), \quad \min_{i \leq k} p_i = \frac{v_k}{W} > \alpha_k \geq \alpha_j \quad (j > k).$$

Enlarge the original control set to caps $a_i = p_i$ on $H = [k]$ and $a_j = \alpha_j$ on its complement. Every original policy is feasible in this relaxation. Theorem 2.3, from zero loads, gives the *joint* bound

$$D \leq \mathbb{E} \max_{i \leq k} N_i, \quad N \sim \text{Mult}(T, p). \quad (1)$$

In particular, no sum of separate head and tail bounds is needed.

Set $\beta = W/(1 + W) \geq 1/2$. Generate the static A_0 -outcomes by independently retaining every reference outcome with probability β , replacing a nonretained outcome by outside. The inside probabilities become $\beta p_i = v_i/(1 + W)$, exactly the static MNL law. Conditional on the reference counts, let I be a deterministic maximum-count index. Its expected retained count is βN_I , so

$$F_T(A_0) \geq \beta \mathbb{E} \max_i N_i \geq \beta D.$$

This expectation argument does not claim that the maximum of the thinned counts equals the thinned maximum on each realization.

If $W = v([n]) < 1$, enlarge the caps instead to $a_i = v_i$. Since their sum is at most one, every nonnegative continuation increment is maximized by simultaneously setting all probabilities to their caps. Thus the constant law with inside probabilities v_i and outside probability $1 - W$ is optimal in this relaxation. Thin its inside outcomes by $\beta = 1/(1 + W) > 1/2$. The resulting inside probabilities are the original full-menu probabilities, and the same conditional-count argument completes the proof. $\square$

Relation to the published proof. The reference prefix, normalization $p_i = v_i/W$, and thinning comparison are already in [1, Section 4.3]. That proof bounds the head and complement separately by $\mathbb{E} \max N_i$ and then uses subadditivity, paying a factor two before thinning. Equation (1) supplies the missing joint comparison. The new upper bound does not contradict the published theorem; it strengthens it. The following construction also sharpens the published lower bound. Neither comparison determines the exact gap or gives a dynamic PTAS.

Corollary 2.4 (A common stochastic reference). *When $v([n]) < 1$, use categorical probabilities $p_i = v_i$ on all products and outside probability $1 - v([n])$. Otherwise use $p_i = v_i/v(A_0)$ on the prefix A_0 above, with no outside mass. Let Z_T be the largest inside count from T iid draws of this law and $\mu_T = \mathbb{E}Z_T$. For every policy and every nondecreasing function f on $\{0, \dots, T\}$,*

$$\mathbb{E}f(M_T^\pi) \leq \mathbb{E}f(Z_T), \quad F_T(A_0) \leq D_T \leq \mu_T \leq 2F_T(A_0) \leq 2D_T.$$

The same categorical law defines Z_r for each $r \leq T$.

Proof. The capped comparison in the preceding proof holds for every nondecreasing terminal transform. Thinning gives $F_T(A_0) \geq \beta\mu_T$ with $\beta \geq 1/2$. Feasibility of the constant menu gives the remaining inequalities. $\square$

2.3 A two-customer improvement to the lower bound

This sharpens the equal-weight, two-customer construction in [1, Section 4.4]. Their Appendix C.2 also identifies two-customer equal-weight instances as large-gap examples. The improvement below comes from the integer choice of static menu size.

Theorem 2.5. *Let $r > 1$ be the unique root of $3r^3 - 3r - 1 = 0$, and set*

$$\Gamma = \frac{(1 + 3r)^2}{6r(1 + r)} = 1.3346234644\dots$$

Then the general adaptivity gap is at least $\Gamma > 1.33462$. An explicit finite instance has $T = 2$, $n = 10000$, and every weight $v = 17/15$:

$$S = \frac{833}{726}, \quad D = \frac{3540766375}{2312408018}, \quad \frac{D}{S} = \frac{75605776125}{56653996441} > \frac{4}{3} + \frac{1}{1000}.$$

Proof. For n equal weights v and two customers, write $q = nv/(1 + nv)$ and $\alpha = v/(1 + v)$. After an inside purchase, the optimal final menu is the purchased singleton, giving conditional payoff $1 + \alpha$. After an outside outcome, offering all products gives payoff q . A first menu of size a therefore gives

$$\frac{q + av(1 + \alpha)}{1 + av}.$$

This increases with a because $1 + \alpha > q$, so offering all products first is optimal and $D = q(2 + \alpha - q)$. Consequently $D \rightarrow 1 + \alpha$ as $n \rightarrow \infty$ with v fixed.

A static menu of size a gives

$$f_v(a) = 1 + \frac{av^2 - 1}{(1 + av)^2}.$$

As a function of real $a \geq 0$, it increases up to $a_* = 1/v + 2/v^2$ and decreases thereafter. At $v = r$, the root equation gives $f_r(2) = f_r(3)$ and $2 < a_* < 3$, so these sizes are optimal for every $n \geq 3$. Hence

$$\lim_{n \rightarrow \infty} \frac{D}{S} = \frac{1 + 2r}{1 + r} \frac{(1 + 3r)^2}{6r(1 + 2r)} = \Gamma.$$

For an exact numerical certificate, the cubic is strictly increasing above one and changes sign between $1137158042603/10^{12}$ and $1137158042604/10^{12}$. The function $R_3(v) = (1 + 3v)^2/(6v(1 + v))$ increases for $v > 1$; substitution of the lower endpoint gives $R_3(v) > 66731/50000 = 1.33462$. Thus the decimal claim requires no floating-point assumption.

For the finite instance, $a_* = 705/289 \in (2, 3)$, while $f_v(3) = 833/726$ exceeds $f_v(2) = 2754/2401$ by $629/1743126$. Substitution in the exact formula for D gives the displayed values and the rational gap inequality. $\square$

3 A dimension-free large-value bound

Theorem 3.1 (Small-menu-family bound). *For every real $0 < \theta \leq 1/4$, a family $\mathcal{F}_\theta$ of at most $g_\theta = \lceil 4/\theta \rceil$ feasible static menus can be constructed from the weights by sorting and grouping. For rational weights and rational θ , the construction uses exact rational arithmetic. Put*

$$R = \max_{A \in \mathcal{F}_\theta} F_T(A), \quad c_\theta = \frac{1+\theta}{1-\theta}, \quad H_\theta = \log g_\theta.$$

Then

$$D \leq c_\theta R + \sqrt{2c_\theta R H_\theta} + H_\theta/3. \quad (2)$$

All logarithms are natural unless a base is displayed. The bound is uniform in n, T , and all positive weights.

Corollary 3.2 (Uniform large-value limit). *As $S \rightarrow \infty$, uniformly over the original model,*

$$D - S = O\left(\sqrt{S \log(2+S)}\right), \quad D/S \rightarrow 1.$$

For $0 < \epsilon < 1$, use $\theta = \epsilon/8$ and define

$$B(\epsilon) = 16\epsilon^{-2} \log(64/\epsilon). \quad (3)$$

If the computed family value $R \geq B(\epsilon)$, its best static menu has value at least $D/(1+\epsilon) \geq (1-\epsilon)D$. Otherwise $D < (1+\epsilon)B(\epsilon)$.

Proof. Use $R \leq S$ in (2), and choose $\theta = S^{-1/2}$ for $S \geq 16$. For the constructive claim, $H_\theta \leq \log(64/\epsilon)$, $c_\theta < 2$, and $c_\theta - 1 \leq 2\epsilon/7$. At $R = B(\epsilon)$, the square-root term divided by R is at most $\epsilon/2$, while $H_\theta/(3R) \leq \epsilon/48$. Their sum with $c_\theta - 1$ is below ϵ . The relative bound improves as R increases. Monotonicity of the right side of (2) gives the complementary assertion. For finite arithmetic, replacing the logarithm in (3) by $\lceil \log_2(64/\epsilon) \rceil$ only increases the threshold and changes no asymptotic order. $\square$

In this section, J_r denotes the capped optimum for the identity terminal payoff and $\alpha_i = v_i/(1+v_i)$.

Lemma 3.3 (Cap cover and saturation). *Let H have caps and loads no smaller than every coordinate outside H . If $\sum_{i \in H} a_i \geq 1$, an optimal capped policy can permanently ignore its complement. If $\sum_{i \in H} a_i = 1$, the fixed iid law $p_i = a_i$ on H is optimal. If instead $\sum_{i=1}^n a_i \leq 1$, the fixed law using every inside probability a_i and the remaining outside mass is optimal, from every state.*

Proof. Let $d_i = J_{r-1}(\ell + e_i) - J_{r-1}(\ell) \geq 0$ be increments of the full capped problem. Lemma 2.2 gives $d_i \geq d_j$ for head i and tail j . Starting from any control, its tail and outside mass can be reassigned to unused head capacity, without decreasing $\sum_i q_i d_i$. There is enough capacity because the head caps sum to at least one. Therefore a head-only control is optimal now. Its transitions preserve head-load dominance, so induction proves permanent exclusion. When head caps sum exactly one, filling them gives the same fixed law at each step. If all caps sum at most one, monotonicity of every increment instead permits filling every cap simultaneously. $\square$

3.1 A dimension-free concentration comparison

Let independent categorical draws have inside counts N_i , with any remaining outside probability. For a group G , set $Y_G = \max_{i \in G} N_i$, taking maximum zero for an empty group. Let $Y_G^{(-t)}$ remove draw t . Pathwise,

$$0 \leq Y_G - Y_G^{(-t)} \leq 1, \quad \sum_t (Y_G - Y_G^{(-t)}) \leq Y_G. \quad (4)$$

Indeed, if there is a unique leading group coordinate, exactly its count many removals reduce the maximum; if leading coordinates tie, no single removal reduces it.

The classical self-bounding inequality [2] gives, with $\mu_G = \mathbb{E}Y_G$,

$$\log \mathbb{E}e^{\lambda(Y_G - \mu_G)} \leq \mu_G(e^\lambda - \lambda - 1) \leq \frac{\mu_G \lambda^2}{2(1 - \lambda/3)}, \quad 0 < \lambda < 3. \quad (5)$$

For completeness, entropy tensorization applied to (4) gives, for $F(\lambda) = \log \mathbb{E}e^{\lambda Y_G}$,

$$\lambda F' - F \leq (e^{-\lambda} + \lambda - 1)F'.$$

Hence $(1 - e^{-\lambda})F' \leq F$, so $F/(e^\lambda - 1)$ is nonincreasing with limit μ_G at zero. This proves the first bound in (5). The second follows by expanding the exponential series. The concentration principle is prior work, not a new probabilistic theorem here.

Lemma 3.4 (Maximum over groups). *For any g groups, which may be dependent through the same categorical draws, let $\mu = \max_G \mathbb{E}Y_G$. Then*

$$\mathbb{E} \max_G Y_G \leq \mu + \sqrt{2\mu \log g} + \frac{1}{3} \log g. \quad (6)$$

The same bound holds for g adaptive counts whose per-round conditional probabilities are all at most a/T , with a replacing μ .

Proof. Log-sum-exp and (5) bound the excess by

$$\frac{\log g}{\lambda} + \frac{\mu \lambda}{2(1 - \lambda/3)}.$$

For $g > 1$ and $\mu > 0$, choose $\lambda = s/(1 + s/3)$ with $s = \sqrt{2 \log(g)/\mu}$. This gives (6); the zero cases follow directly. No independence between groups is used. For an adaptive count N , iterated conditioning gives $\mathbb{E}e^{\lambda N} \leq \exp(a(e^\lambda - 1))$, so the identical log-sum argument applies centered at a . $\square$

3.2 Constructing the physical menu family

Sort $v_1 \geq \dots \geq v_n$, equivalently $\alpha_1 \geq \dots \geq \alpha_n$. There are three cases.

Case I: $\sum_i \alpha_i \leq 1$. Use the iid reference probabilities $p_i = \alpha_i$, with the remaining outside mass. Lemma 3.3 bounds D by its expected maximum. Let the reference head be all products and put $B = 1$.

Case II: a small cap cover. If the cap sum exceeds one, let k be the shortest prefix with $\sum_{i \leq k} \alpha_i \geq 1$. When $k < 1/\theta$, Lemma 3.3 removes the tail in the capped upper model. Every remaining adaptive count has conditional probability at most α_1 . With $a = T\alpha_1$, Lemma 3.4 gives

$$D \leq a + \sqrt{2a \log k} + \frac{1}{3} \log k.$$

Choose the physical family consisting only of singleton $\{1\}$, whose value is exactly $R = a$. Since $k \leq g_\theta$ and $c_\theta \geq 1$, this proves (2) in this case.

Case III: a large cap cover. When $k \geq 1/\theta$, use head $H = [k - 1]$ and

$$B = \sum_{i < k} \alpha_i < 1, \quad p_i = \alpha_i / B \quad (i \in H).$$

Minimality and order imply

$$1 \leq B + \alpha_k \leq B + B/(k - 1), \quad B \geq 1 - 1/k \geq 1 - \theta.$$

Increase head caps to p_i and leave tail caps unchanged. Every head cap is at least every tail cap. By saturation, the iid law p on H dominates the complete original adaptive optimum.

Grouping for Cases I and III. Every head item with $v_i > \theta$ forms a singleton group. Pack the remaining items, in order, by next-fit into groups of total weight at most θ . The head cap sum is at most one, so there are fewer than $(1 + \theta)/\theta$ heavy items and the light items have total weight at most $1 + \theta$. Consecutive light groups together exceed θ , giving at most $2(1 + \theta)/\theta + 1$ light groups. Thus the number g of groups is at most

$$3/\theta + 4 \leq 4/\theta \leq g_\theta.$$

Every group is an actual feasible static menu. Set $\beta = B/(1 + \theta) \geq 1/c_\theta$. For singleton heavy groups, physical purchase probability is $\alpha_i = Bp_i \geq \beta p_i$. For a light group G ,

$$\frac{v_i}{1 + v(G)} \geq \frac{\alpha_i}{1 + \theta} = \beta p_i.$$

Let Y_G be the group's maximum count in the common reference process and $\mu_G = \mathbb{E}Y_G$. Independently thin each reference draw with probability β , then fill residual probability to reach the physical group's probabilities. Conditional on the complete reference path, the retained count of a reference-maximizing group item has mean βY_G . The actual group maximum dominates it. Hence

$$F_T(G) \geq \beta \mu_G, \quad \mu := \max_G \mu_G \leq c_\theta R. \quad (7)$$

The reference groups cover every reference inside coordinate, so

$$D \leq \mathbb{E} \max_G Y_G \leq \mu + \sqrt{2\mu \log g} + \frac{1}{3} \log g.$$

Apply (7). This completes Theorem 3.1.

The family has $O(1/\theta)$ menus even if the number of head products is very large. These menus need not be weight prefixes; the grouping is essential. Their individual values can be computed by the established static evaluator [1]. The argument does not depend on the static-prefix theorem in Appendix C.

4 Exponential tails and truncation

4.1 Uniform upper tails and an exact truncation certificate

Theorem 4.1 (Exponential-moment envelope). *For every original adaptive policy and every $\lambda \geq 0$,*

$$\mathbb{E}e^{\lambda M_T^\pi} \leq \exp\{\mu_T(e^\lambda - 1)\} \leq \exp\{2D_T(e^\lambda - 1)\}. \quad (8)$$

Consequently, for $x \geq 0$,

$$\mathbb{P}(M_T^\pi \geq \mu_T + x) \leq \exp\left(-\frac{x^2}{2(\mu_T + x/3)}\right).$$

Proof. For the iid reference maximum $Z = Z_T$, with $\mu = \mu_T$, delete one draw and denote the result by $Z^{(-t)}$. Pathwise,

$$0 \leq Z - Z^{(-t)} \leq 1, \quad \sum_t (Z - Z^{(-t)}) \leq Z.$$

When a positive leader is unique, precisely its count many deletions reduce the maximum; with tied leaders none does. At zero the sum is zero. The classical self-bounding inequality [2] gives $\log \mathbb{E}e^{\lambda(Z-\mu)} \leq \mu(e^\lambda - \lambda - 1)$. Corollary 2.4 transfers its positive exponential moments to the adaptive process. The stated Bernstein form follows by the usual exponential Markov optimization, using $e^\lambda - \lambda - 1 \leq \lambda^2/[2(1 - \lambda/3)]$ for $0 \leq \lambda < 3$. $\square$

We do *not* assert that the adaptive maximum is itself a self-bounding function of independent draws. Nor do we center this bound at $\mathbb{E}M_T^\pi$ or claim stochastic domination by a Poisson distribution. The center/proxy is the common reference mean μ_T .

After arbitrary histories. If τ is a stopping time, conditional on its complete history, then

$$M_T - M_\tau \leq \max_i (L_i(T) - L_i(\tau)).$$

The future choice rule can be replayed from zero by storing the original initial offset as internal history. Thus its future-count maximum is dominated by $Z_{T-\tau}$. In particular, the conditional exponential moment of the maximum increase is at most $\exp\{\mu_{T-\tau}(e^\lambda - 1)\}$.

Let $D_{T,h} = \sup_\pi \mathbb{E} \min(M_T^\pi, h)$. It is attained by a policy offering nothing once the maximum reaches h .

Theorem 4.2 (Reference-tail and logarithmic-error cutoff). *For every nonnegative integer h ,*

$$0 \leq D_T - D_{T,h} \leq E_h := \mathbb{E}(Z_T - h)_+ \leq \frac{e^{\mu_T} - 1}{2^{h+1} - 1}. \quad (9)$$

If $D_T \leq U$ and $0 < \gamma < 1$, stopping at

$$h = \min\left\{T, \left\lceil \frac{2U + \log(2/\gamma)}{\log 2} \right\rceil\right\} \quad (10)$$

loses at most γD_T . The fully rational conservative choice $h = \min\{T, \lceil 4U \rceil + \lceil \log_2(2/\gamma) \rceil\}$ also works.

Proof. Cap an optimal original policy; the loss is at most $\mathbb{E}(M_T^\pi - h)_+$, which Corollary 2.4 bounds by E_h . For integers $z, h \geq 0$,

$$(2^{h+1} - 1)(z - h)_+ \leq 2^z - 1.$$

For $z = h + q > h$, subtracting the left side gives $2^h(2^q - 2q) + q - 1 \geq 0$. Take expectations and apply Theorem 4.1 to Z_T at $\lambda = \log 2$. Since $e^{\mu_T} - 1 \leq \mu_T e^{\mu_T}$ and $\mu_T \leq 2D_T \leq 2U$, a denominator of at least $2e^{\mu_T}/\gamma$ suffices. Formula (10) ensures this unless $h = T$, in which case the loss is zero. The rational substitute uses $2^{4U} \geq e^{2U}$. $\square$

E_h is an exact rational quantity for rational weights, computable from capped-count generating functions in $O(nT^3)$ arithmetic operations. A particularly useful finite certificate chooses the smallest h with $E_h \leq \gamma F_T(A_0)$. Solving only the stopped problem then yields the rigorous interval $[D_{T,h}, D_{T,h} + E_h]$ and a $(1 - \gamma)$ -approximate physical policy. That stopped solve can still have a combinatorial state space.

A separate renewal argument without the capped comparison

Let $q_T(k) = \sup_\pi \mathbb{P}(M_T^\pi \geq k)$. First hitting level a occurs with maximum exactly a . To attain $a + b$, some subsequent inside count must increase by at least b . Conditioning at the hitting time and replaying the continuation from zero proves

$$q_T(a + b) \leq q_T(a)q_T(b). \quad (11)$$

Thus $q_T(kb) \leq (D_T/b)^k$ by Markov's inequality. The conditional expected increase after hitting any level is at most D_T , so $D_T - D_{T,kb} \leq D_T(D_T/b)^k$. This independently gives an $O(U \log(1/\gamma))$ cutoff. The stronger $O(U + \log(1/\gamma))$ cutoff above uses the stochastic reference and classical self-bounding input.

5 A smaller quasi-polynomial approximation scheme

Lemma 5.1 (Deleting light products and lifting nearby weights). *Let $\alpha = \max_i v_i/(1 + v_i)$ and $0 < \gamma < 1/2$. For a fixed integer $h \geq 1$, the capped payoff is $\min\{\max_i L_i, h\}$. Deleting products with $v_i \leq \gamma\alpha/n$ loses at most γD in capped or uncapped value from zero. On any retained universe, suppose*

$$v_i/(1 + \gamma) \leq \tilde{v}_i \leq v_i.$$

For every adaptive policy under either weight vector there is a causal policy under the other with at least $1/(1 + \gamma)$ of its expected maximum, also for the capped payoff.

Proof. Deleting products from each requested menu increases every retained product's probability. Couple real and virtual choices so each virtual retained purchase is also real, and drive future menus by virtual history. Any remaining possible maximum advantage is at most the total number of virtual deleted purchases. Its expectation is at most $T \sum_{\text{deleted}} v_i \leq \gamma T \alpha \leq \gamma D$.

For nearby weights, every inside probability for the same menu is at least $1/(1 + \gamma)$ of its counterpart, in both directions. For two categorical laws p, q , the coupling with diagonal $\min(p_i, q_i)$ and product residuals has the prescribed marginals and conditional match probability at least $1/(1 + \gamma)$ on every virtual inside event. A real controller implements it by sampling the virtual observation conditionally on the current real one; future menus depend only on virtual history.

Equivalently, generate the entire virtual path first and then use the corresponding coupling kernels. Conditional on that virtual path, select a terminal winning product and its first $\min\{h, L_i\}$

purchase events. Their expected matched count is at least $1/(1+\gamma)$ of the virtual capped maximum. The real capped maximum is at least this matched count. This proves the claim without conditioning independent observations on a future success event. This policy-lifting idea is established in [1, Appendix D.3]; the capped version follows from the same designated-event argument. $\square$

Theorem 5.2. *For rational weights and rational $0 < \epsilon < 1/2$, the original Dynamic-MLA problem admits a $(1 - \epsilon)$ approximate policy whose preprocessing and per-customer computation are bounded by*

$$\text{poly}(\text{input bit length})(n+1)^{O(\epsilon^{-3} \log(1/\epsilon) \log(n/\epsilon))}.$$

For fixed ϵ , this is $n^{O_\epsilon(\log n)}$. The theorem does not give polynomial time in n at fixed ϵ .

Proof. Put $\gamma = \epsilon/8$. First process three easy cases, before any horizon-length dynamic program. If $v_{\max} \geq 1/\gamma$, offering that product has value $T\alpha \geq T/(1+\gamma) \geq (1-\gamma)D$. If $v([n]) \leq \gamma$, the reference law with probabilities v_i and outside remainder is optimal in its capped relaxation; thinning by $1/(1+v([n]))$ proves the full menu is at least $(1-\gamma)D$.

Put $\Lambda_n = \lceil \log_2(2n) \rceil$, an exactly computable integer upper bound for $\log(2n)$. Finally, if $a = T\alpha \geq 16\gamma^{-2}\Lambda_n$, the adaptive-count part of Lemma 3.4, applied to all n products, gives

$$D \leq a + \sqrt{2a\Lambda_n} + \frac{1}{3}\Lambda_n \leq (1+\gamma)a.$$

The heaviest singleton is again sufficient. If none of these tests succeeds, $v([n]) > \gamma$ implies $\alpha \geq \gamma/(n+\gamma)$, whence

$$T < 16\gamma^{-3}(n+\gamma)\Lambda_n. \tag{12}$$

Thus subsequent exact static value computations are polynomial in n at fixed accuracy. Rational conservative upper bounds for logarithms make these tests implementable; strict real-log comparisons are unnecessary.

Construct Theorem 3.1's family at $\theta = \gamma/8$ and compute its best value R . Use the rational threshold $\hat{B}(\gamma) = 16\gamma^{-2}\lceil \log_2(64/\gamma) \rceil \geq B(\gamma)$. If $R \geq \hat{B}(\gamma)$, Corollary 3.2 finishes. Otherwise the same corollary's proof gives

$$D \leq U := (1+\gamma)\hat{B}(\gamma) = O(\gamma^{-2} \log(1/\gamma)).$$

Set $h = \min\{T, \lceil 4U \rceil + \lceil \log_2(2/\gamma) \rceil\} = O(\gamma^{-2} \log(1/\gamma))$. Theorem 4.2 loses at most γD . Delete the light products of Lemma 5.1, losing at most another γD . At least a heaviest product remains.

Round each retained weight down to the grid $v_{\max}(1+\gamma)^{-j}$, $j \geq 0$. Since $v_{\max} < 1/\gamma$ and retained weights exceed $\gamma\alpha/n$,

$$\frac{v_{\max}}{\min_{\text{retained}} v_i} < \frac{n(1+v_{\max})}{\gamma} \leq \frac{2n}{\gamma^2}.$$

The number of distinct rounded weights is therefore $J = O(\gamma^{-1} \log(n/\gamma))$. Lemma 5.1 shows that the optimum of the rounded, capped instance is at least $(1-2\gamma)D/(1+\gamma)$.

Compute it exactly by recording, for every rounded weight class and every load $0, \dots, h-1$, the number of products in that cell. Stop at load h . There are at most $(n+1)^{Jh}$ histogram states at each remaining horizon. A cell's members have identical continuation increments. At a state, sorting these increments solves the ordinary MNL artificial-revenue subproblem; a maximizing menu includes whole cells, with at most Jh prefix candidates. Thus each transition and action optimization has polynomial overhead. This histogram formulation, weight rounding, and action oracle follow the source's framework [1, Section 5 and Appendix D.5]; the new value ceiling removes two powers of $\log n$ from the exponent.

Lift the resulting policy back to the original weights using Lemma 5.1. It achieves at least

$$\frac{1 - 2\gamma}{(1 + \gamma)^2} D \geq (1 - 4\gamma) D \geq (1 - \epsilon) D.$$

Since $Jh = O(\gamma^{-3} \log(1/\gamma) \log(n/\gamma))$, this proves the complexity claim. Equation (12) ensures that evaluating R did not assume the bounded-value conclusion before establishing it. In the residual branch one additionally has $T \leq U/\alpha = O_\epsilon(n)$.

For bit complexity, the rounded rational weights have bit length polynomial in the rational input length and J . Clear their denominators once, writing class weights as a_j/B with positive integers a_j, B . A menu containing k_j products of class j has transition denominator $d(k) = B + \sum_j k_j a_j$. There are at most $K = (n + 1)^J$ such count vectors, and each $d(k)$ has at most b bits for a polynomially bounded b . Let C be their product. Every deterministic finite-horizon policy value for the capped integer payoff has denominator dividing C^T and numerator at most hC^T . Bellman maximization selects one of these values, so its bit length is $O(TKb + \log h)$. Polynomial-time integer arithmetic and exact comparisons therefore preserve the displayed $(n + 1)^{O(Jh)}$ order. The static-value branch uses only polynomial-bit operations after the horizon reduction. Per-customer policy lifting adds polynomial overhead; an exponentially large input horizon is not unrolled in the easy branches. $\square$

The unresolved step. The load ceiling is now independent of n , but the number of weight classes remains $\Theta_\epsilon(\log n)$. The histogram state count can still be $n^{\Theta_\epsilon(\log n)}$. No theorem here compresses all class-by-load counts to $n^{O_\epsilon(1)}$. Consequently, neither a general adaptive PTAS nor exact adaptive optimization nor the sharp constant adaptivity gap is established. Our lower and upper bounds bracket that constant by $[\Gamma, 2]$, with Γ from Theorem 2.5. The large-value conclusion is that any fixed nonvanishing relative gain from adaptation must occur at bounded expected value.

6 Limits of small lists of purchased candidates

6.1 A bounded-value obstruction to small candidate lists

A small maximum does not imply that only a bounded number of purchased products matter. Here is a quantitative statement about menus, not memory.

Theorem 6.1 (Previously purchased candidates can be numerous). *For each integer $T \geq 20$, let $n = T^2$ and $v_i = 1/n$ for every product. Then*

$$\frac{11}{10} < D_T < \frac{3}{2}.$$

If a policy offers at most K previously purchased products on every history at every round, then

$$\mathbb{E} M_T^\pi \leq 1 + \frac{KT}{n+1} \leq 1 + \frac{K}{T}. \quad (13)$$

In particular, if $K \leq T/25$, it is not a 95% approximation.

Proof. Let B_t be the number of positive-load products in the current menu, and let R_t indicate a purchase of an already-purchased product. Conditional on the history and menu,

$$\mathbb{P}(R_t = 1 \mid \mathcal{F}_{t-1}, S_t) = \frac{B_t}{n + |S_t|} \leq \frac{B_t}{n + 1}.$$

Pathwise, $M_T \leq 1 + \sum_t R_t$. As $B_t \leq t-1$, every adaptive policy has value at most $1 + \binom{T}{2}/(n+1) < 3/2$. Under the additional bound $B_t \leq K$, equation (13) follows.

For a lower bound offer all products statically. Each inside probability is $1/(2n)$ and the outside probability is $1/2$. Let C count pairs of customer times choosing the same inside product. Exact counting gives

$$\mathbb{E}C = \frac{\binom{T}{2}}{4n}, \quad \mathbb{E}C^2 = \mathbb{E}C + \frac{6\binom{T}{3}}{8n^2} + \frac{6\binom{T}{4}}{16n^2}.$$

Therefore, by Cauchy–Schwarz,

$$\mathbb{P}(C > 0) \geq \frac{(\mathbb{E}C)^2}{\mathbb{E}C^2} = \frac{T(T-1)}{9T^2 + 3T - 10}.$$

The expected maximum is at least the probability of one inside purchase plus the probability of a repeated inside purchase, hence

$$D_T \geq 1 - 2^{-T} + \frac{T(T-1)}{9T^2 + 3T - 10} > \frac{11}{10}. \quad (14)$$

The rational fraction is increasing for $T \geq 2$; its derivative has numerator $2(6T^2 - 10T + 5) > 0$. Check the last strict inequality at $T = 20$. If $K \leq T/25$, the restricted value is at most $26/25 = 1.04$, whereas $(19/20)D_T > 209/200 = 1.045$. $\square$

More generally any near-optimal policy must have large aggregate exposure:

$$\mathbb{E} \sum_{t=1}^T B_t \geq (n+1)(\mathbb{E}M_T^\pi - 1).$$

Thus a fixed positive improvement above one requires average B_t of order T . All these instances lie in the bounded-value regime, despite unbounded n and T .

This does *not* prove a memory lower bound, hardness, or impossibility of histogram compression. A static full menu does not need to remember labels at all. The theorem rules out physically excluding all but an accuracy-dependent number of already-purchased competitors. Large groups might still be summarized rather than discarded.

6.2 Exact aggregate computation in the stress-test family

The preceding distinction can be demonstrated constructively. For n equal weights v , cap the payoff at two. Before a repeat, the only needed state is s , the number of distinct products purchased. For a restriction of at most K purchased products offered, let $b = \min(s, K)$; in the unrestricted case let $b = s$.

With r customers left, write $C_r(s)$ for the capped value. Then $C_0(s) = \mathbf{1}\{s > 0\}$, and

$$C_r(s) = \max \left\{ \frac{C_{r-1}(s) + 2vb}{1 + vb}, \quad \frac{C_{r-1}(s) + 2vb + v(n-s)C_{r-1}(s+1)}{1 + v(b+n-s)} \right\}. \quad (15)$$

Omit the second alternative when $s = n$. A repeat immediately gives payoff two, so all permitted purchased products can be offered. With their count fixed, the value is fractional-linear in the number of offered unpurchased products; one endpoint, zero or all $n-s$, is optimal. This proves the recursion rather than assuming the endpoint policy. It uses $O(T^2)$ states.

To relate it to the full uncapped problem, a general cap argument gives

$$\mathbb{E} \sum_i \binom{L_i(T)}{k} \leq \binom{T}{k} \alpha_{\max}^{k-1}, \quad \alpha_{\max} = \max_i \frac{v_i}{1 + v_i}. \quad (16)$$

For each k -tuple of times, peel off the final conditional selection probability repeatedly and then sum the first inside outcome. Independence of actual counts is unnecessary. Since $(M - 2)_+ \leq \sum_i \binom{L_i}{3}$,

$$C_T(0) \leq D_T \leq C_T(0) + \frac{\binom{T}{3}}{(T^2 + 1)^2}$$

in Theorem 6.1's family.

For large T , the code rounds every Bellman value down to a multiple of 2^{-50} . Monotonicity and sup-norm nonexpansiveness of the Bellman operator give total value error at most $T2^{-50}$. Thus the displayed intervals below have rigorous integer-arithmetic endpoints, not Monte Carlo uncertainty:

T	$n = T^2$	lower bound on D_T	upper bound on D_T
40	1,600	1.15127635	1.15513092
160	25,600	1.15296176	1.15398390
640	409,600	1.15337939	1.15363860

The decimal endpoints here are rounded outward. The $O(T^2)$ computation is for this equal-weight capped laboratory, not a solution for heterogeneous weights. It illustrates precisely why aggregate state information can work where a constant list of retained candidates cannot.

7 Verification and scope

The accompanying research code uses exact integers and rational fractions for finite Bellman values, multinomial distributions, coefficient checks, couplings, and pruning certificates. Separate all-menu or direct count-state references check the specialized evaluators. Numerical diagnostics involving logarithms and exponentials use 80-digit arithmetic; they are not interval proofs. The large equal-weight intervals in Section 6 instead use exact integer rounding with a proved error bound. These tests are reproducibility diagnostics, with overlapping instances, and do not replace the analytic arguments. The supplied code contains the menu construction, exact reference solvers, histogram core, and coupling kernels; it is not an optimized end-to-end implementation of the full QPTAS. The general adaptive PTAS and sharp gap remain open in this work. Structural statements here do not extend to mixed MNL populations, inventory constraints, or weighted load objectives.

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A Continuation-state dominance and pruning

Let $V_r(\ell)$ denote the optimal terminal maximum from current integer loads ℓ and r remaining customers. The known Bellman recursion is

$$V_r(\ell) = V_{r-1}(\ell) + \max_{S \subseteq [n]} \frac{\sum_{i \in S} v_i d_i}{1 + \sum_{i \in S} v_i}, \quad d_i = V_{r-1}(\ell + e_i) - V_{r-1}(\ell), \quad (17)$$

with $V_0(\ell) = \max_i \ell_i$. The inner maximization is the usual MNL assortment oracle, sorted by d_i , not by v_i . Its tractability does not remove the combinatorial state space. The following reductions concern continuation states and do not imply a polynomial-size state space.

A.1 Continuation increments: coverage and catch-up tails

Put $M = \max_i \ell_i$ and $\alpha_i = v_i/(1 + v_i)$. For $r \geq 1$ define

$$b_i(r, \ell) = \mathbb{P}\{\text{Bin}(r - 1, \alpha_i) \geq M - \ell_i\}. \quad (18)$$

A nonpositive threshold gives probability one, and a threshold exceeding $r - 1$ gives zero. These are individual comparison binomials; no independence between actual product loads is asserted.

Lemma A.1. *At every original-model state,*

$$0 \leq d_i \leq b_i(r, \ell) \leq 1, \quad \sum_{i=1}^n d_i \geq 1.$$

Proof. Any policy can be written as a map from observed outcome histories to menus. Use the same map and the same outcome sequence from two initial load vectors. Increasing initial coordinate i by one changes the terminal maximum by one precisely when i is a weak terminal leader in the unshifted vector. This statement includes ties and the all-outside outcome.

For the lower bound on the sum, fix a policy optimal at ℓ with $r - 1$ customers remaining and replay it from each $\ell + e_i$. It follows that d_i is at least that policy's weak-leader probability for i . Every terminal vector has at least one weak leader, so their sum is at least one. Notice that this uses a single common policy, not different optimal policies whose probabilities could not be added.

For the upper bound, instead fix a policy optimal at $\ell + e_i$ and replay it from ℓ . Its payoff difference bounds d_i above. For i to be a weak terminal leader in the unshifted vector it must collect at least $M - \ell_i$ new purchases. At every round, under any history-dependent menu, its conditional selection probability is at most α_i . Successively using fresh uniform random variables couples its count below $\text{Bin}(r - 1, \alpha_i)$. This proves the tail bound. Monotonicity and one-Lipschitzness in an individual initial load give $0 \leq d_i \leq 1$ as well. $\square$

A.2 A permanent zero-loss deletion certificate

Let E be proposed exclusions and $A = [n] \setminus E$ the nonempty retained set. Define

$$B_E = \sum_{i \in E} b_i(r, \ell), \quad H_A = \sum_{i \in A} \alpha_i^{-1}, \quad \lambda_A = \frac{1 - B_E}{H_A}. \quad (19)$$

Theorem A.2 (Permanent pruning). *If $E \neq \emptyset$ and*

$$B_E < 1, \quad \max_{i \in E} b_i(r, \ell) < \lambda_A, \quad (20)$$

then every optimal action excludes E , at this state and at every subsequent state reachable while E remains unoffered. Hence the original optimal value is exactly the optimal value of the subproblem retaining only A .

Proof. Let ρ denote the maximized ratio in (17). Singletons are feasible, so $\rho \geq \alpha_i d_i$ for every i . Lemma A.1 implies

$$1 - B_E \leq \sum_{i \in A} d_i \leq \rho \sum_{i \in A} \alpha_i^{-1},$$

hence $\rho \geq \lambda_A$. Every $i \in E$ has $d_i \leq b_i < \lambda_A \leq \rho$. If an optimal menu contained such an i , deleting it would increase its ratio by $v_i(\rho - d_i)/(1 + \sum_{j \in S \setminus \{i\}} v_j) > 0$, a contradiction.

Now consider any later state while products in E remain unoffered. Their initial loads are unchanged, the maximum load can only increase, and the remaining horizon can only decrease. Therefore $b'_i \leq b_i$ for $i \in E$. The same retained set satisfies $B'_E \leq B_E$ and $\lambda'_A \geq \lambda_A$. The strict exclusion condition persists. Backward induction establishes an optimal policy using only A ; in fact every optimal action on its reached histories avoids E .

No initial maximum is lost on deletion. A current leader has $b_i = 1$ and cannot belong to a set with $B_E < 1$. Thus A contains every current leader, and the future terminal maximum of retained products is always at least every fixed initial excluded load. The ordinary reduced objective is therefore the right one. $\square$

The criterion is computable without any Bellman value. Evaluate the binomial tails, sort them increasingly, and test the at most $n - 1$ candidate prefixes. Prefix sums of b_i and suffix sums of α_i^{-1} make the scan linear after sorting. This finds a largest certifiable exclusion set: if a set passes with maximum tail t , any unexcluded item with $b_i \leq t < \lambda_A$ can be added to it; λ_A increases since $b_i < \lambda_A/\alpha_i$. Repeating produces the full sorted prefix at that threshold. A current leader remains.

Exact 40-product example. Take

$$(v_1, \dots, v_{40}) = (2, 1, 1/2, 1/3, \dots, 1/39), \quad \ell = (8, 7, 0, \dots, 0), \quad r = 10.$$

For $E = \{3, \dots, 40\}$, exact arithmetic gives

$$B_E < 1/500, \quad \max_{i \in E} b_i = 19/19683 < 1/1000, \quad H_A = 3/2 + 2 = 7/2.$$

Therefore $\lambda_A > 499/1750 > 1/1000$: all 38 products can be permanently removed, despite their positive chance of catching up if offered. The exact retained two-product solver gives $V_{10}(\ell) = 44/3$ with first menu $\{1\}$. The full 40-label Bellman table was not enumerated. This state is reachable with positive probability after 15 purchases in the original process.

A.3 Load-weight dominance and the first adaptive menu

We next compare states, retaining arbitrary weights and horizons.

Lemma A.3 (Feasible one-step probability vectors). *Allowing randomized feasible assortments, the attainable next-choice vectors are*

$$\mathcal{P}_v = \{p \in \mathbb{R}_+^{n+1} : \sum_{i=0}^n p_i = 1, \ p_i \leq v_i p_0 \ (i \geq 1)\}.$$

Proof. Every deterministic MNL vector satisfies the inequalities, and so does its convex hull. Conversely, $p_0 > 0$ for every vector in this polytope. Put $z_i = p_i/(v_i p_0) \in [0, 1]$ and choose any distribution γ of menus with inclusion marginals z_i , for example independent Bernoulli inclusion. Define

$$\lambda_S = p_0 \left(1 + \sum_{i \in S} v_i \right) \gamma_S.$$

Its total mass is $p_0 + \sum_i p_i = 1$. Offering according to λ induces p exactly. The Bernoulli law γ is a reweighted auxiliary distribution, not in general the physical inclusion law λ . $\square$

Theorem A.4 (Heavier concentrated pair). *Suppose $v_i \geq v_j$. States x, y agree outside $\{i, j\}$, satisfy $x_i + x_j = y_i + y_j$, and obey $x_i \geq \max(y_i, y_j)$. Then*

$$V_r(x) \geq V_r(y) \quad \text{for every } r.$$

More strongly, any policy from y can be coupled to a feasible randomized policy from x whose maximum load is at least as large at every time. Thus the comparison also holds for optimal probabilities of exceeding any terminal maximum-load threshold.

Proof. Maintain either equality of the states or the displayed pair-concentration relation. If the states coincide, use identical choices. If $x_i > \max(y_i, y_j)$, again use the same choice vector and the same outcome. All loads are integer, so the leader has at least a one-unit margin over both y coordinates and at least a two-unit margin over x_j ; the relation persists.

The only nonidentical boundary case is a swapped pair: $x_i = y_j > x_j = y_i$. Let q be the next-choice vector used by the policy from y . Retain every other probability, including $p_0 = q_0$, and set

$$p_j = \min\{q_i, v_j q_0\}, \quad p_i = q_i + q_j - p_j.$$

We have $p_j \leq v_j q_0$, $p_i \geq q_j$, and $p_i \leq v_i q_0$. For the last inequality, if $q_i \leq v_j q_0$, then $p_i = q_j \leq v_j q_0 \leq v_i q_0$; otherwise $p_i = q_i + q_j - v_j q_0 \leq q_i \leq v_i q_0$. Lemma A.3 makes p a physically feasible menu lottery.

Couple a j outcome from y with an i outcome from x (mass q_j). Couple an i outcome from y to j from x with mass p_j , and to i from x with mass $q_i - p_j$. Other outcomes match. Every resulting pair preserves the relation or coincides. This includes the case where a one-unit leader is caught and both pairs become tied. Iterate the construction. The policy from x can internally maintain the coupled virtual history from y , so it is nonanticipating. Randomization does not improve the optimal finite-horizon value beyond a deterministic Bellman policy. $\square$

Corollary A.5. *If $v_i \geq v_j$ and $\ell_i \geq \ell_j$, then $d_i \geq d_j$. Consequently:*

1. *At zero initial loads, an optimal first menu is a raw-weight prefix for every horizon.*
2. *At any state, a maximal optimal menu can be chosen upward closed in the partial order defined by simultaneous weight and load dominance.*
3. *Within one current-load level, that menu selects a prefix by weight; within one equal-weight class, it selects a prefix by current load.*

Proof. Apply Theorem A.4 to $x = \ell + e_i$, $y = \ell + e_j$ and horizon $r - 1$. For (17), an optimal ratio ρ permits exactly all items with $d_i \geq \rho$ to be included (equality items can all be included without changing the ratio). This menu is upward closed. At the initial state all loads are equal, so the order is by weight alone. $\square$

The first-menu claim is not a theorem that all subsequent menus are raw-weight prefixes. For example, $(v_1, v_2) = (2, 1)$, $T = 2$ requires offering only product 2 after it is the first purchase. Nor is a current-leaders-only policy optimal: with two weights $1/100$, state $(1, 0)$ and three remaining customers, the unique optimal menu is $\{1, 2\}$.

A.4 A quantitative bound when exact pruning is unavailable

Let $V_r^A(\ell)$ denote the original objective when only products in A may be offered. Initial excluded loads are still present; equivalently retain $M = \max_i \ell_i$ as a fixed terminal baseline. If A contains a current leader, an ordinary retained-coordinate solver already preserves this baseline. Put $g_i = M - \ell_i$ and

$$\mathcal{E}_E(r, \ell) = \sum_{k=1}^r \min \left\{ 1, \sum_{i \in E} \mathbb{P}\{\text{Bin}(r, \alpha_i) \geq g_i + k\} \right\}.$$

Theorem A.6 (Certified omission loss). *For every $A \subseteq [n]$, with $E = [n] \setminus A$,*

$$V_r^A(\ell) \leq V_r(\ell) \leq V_r^A(\ell) + \mathcal{E}_E(r, \ell).$$

Proof. Simulate an optimal original policy virtually. When it requests S , physically offer $S \cap A$. Let $d_A = 1 + \sum_{i \in S \cap A} v_i$ and $d_E = \sum_{i \in S \cap E} v_i$. After the real choice, retain it as the virtual choice with probability $d_A/(d_A + d_E)$. Otherwise generate a virtual excluded choice proportional to its weight. This gives the original MNL law, with every virtual retained purchase also a real retained purchase. The real policy uses only feasible menus and internally generated randomness, never future data. Its retained terminal maximum with baseline M dominates the virtual one.

On every virtual path, the excess of the full maximum over that baseline and retained maximum is at most $\max_{i \in E} (\ell_i + N_i - M)_+$. The adaptive count N_i is stochastically bounded by $\text{Bin}(r, \alpha_i)$. Apply the union bound at each integer overshoot level, clipped to one. Its expectation is at most $\mathcal{E}_E$. Taking expectations and using the optimality of V_r^A proves the result. No independence between product counts is used. $\square$

Solving a small retained subproblem thus gives an explicit interval for the original value. In particular $\mathcal{E}_E \leq \varepsilon V_r^A$ certifies $V_r^A \geq V_r/(1 + \varepsilon) \geq (1 - \varepsilon)V_r$. No theorem here guarantees that a small retained set exists at every original state.

B Computable continuation upper bounds

The comparison also yields upper certificates when the actual loads do not satisfy head dominance. Let $H \neq \emptyset$, choose p with

$$\sum_{i \in H} p_i = 1, \quad p_i \geq \alpha_i \ (i \in H), \quad \min_{i \in H} p_i \geq \max_{j \notin H} \alpha_j,$$

and put $m_E = \max_{j \notin H} \ell_j$ (zero for an empty complement) and $\ell_i^+ = \max(\ell_i, m_E)$ for $i \in H$. Raise those head loads in the comparison system; monotonicity and Theorem 2.3 give

$$V_r^{\text{MNL}}(\ell) \leq \mathbb{E} \max_{i \in H} (\ell_i^+ + N_i), \quad N \sim \text{Mult}(r, p). \quad (21)$$

The head probabilities are artificial upper-bound probabilities, not a menu recommended to the real controller.

One explicit family of certificates sorts the singleton caps decreasingly. For each head prefix with $\sum_{i \in H} \alpha_i \leq 1$, choose

$$p_i = \alpha_i + \frac{1 - \sum_{j \in H} \alpha_j}{|H|}.$$

Then $\min_H p_i \geq \min_H \alpha_i \geq \max_E \alpha_j$. If all caps sum to at most one, the constant law with inside probabilities α_i and residual outside mass is an additional, tighter candidate without raised loads.

For rational probabilities the upper values in (21) are exactly evaluable in $O(|H|r^3)$ integer arithmetic operations by a capped-count convolution. Clear denominators, obtaining integer weights $a_0, a_1, \dots, a_k$ with total B . At a candidate maximum h , initialize $Q_0(t) = a_0^t$ and set

$$Q_i(t) = \sum_{s=0}^{\min(t, h-\ell_i^+)} \binom{t}{s} a_i^s Q_{i-1}(t-s).$$

Then $Q_k(r)/B^r$ is the probability that the maximum is at most h . Only r thresholds, from the largest initial load to one below it plus r , are needed. Thus huge binary-encoded initial loads do not add an iteration over their magnitude. Intermediate integers have polynomial bit length in numerical r and input size. The supplied code evaluates a minimum over these certificates, not a general optimal adaptive policy.

C An alternative static-prefix proof

The static top- k conclusion below is proved by Zhao and Fu [6, Theorem 4.1]. Their Proposition 4.1 supplies the same one-product quasiconvexity used in a deletion-or-upgrade exchange. We give an alternative route through a coefficientwise comparison of pairwise weak-leader odds, rather than their distributional and head-start argument. The coefficientwise statement is not stated in [6]; this comparison alone does not establish its priority in the wider literature. For the fixed-menu objective F_T defined in Section 1, sort the weights in decreasing order.

Theorem C.1 (Weight-ordered optimality). *For every finite $T \geq 1$, every optimal assortment has no strict weight inversion: it cannot include i and exclude j when $v_j > v_i$. Consequently, after sorting $v_1 \geq \dots \geq v_n$, some optimum is a prefix $\{1, \dots, k\}$. Equal-weight products may be ordered arbitrarily. An optimum and its exact rational objective can be computed in $O(nT^3)$ integer arithmetic operations after sorting, with polynomial bit complexity in n, T , and the rational input length.*

The structural statement follows from the strict endpoint property

$$F_T(C \cup \{a\}) < \max\{F_T(C), F_T(C \cup \{b\})\} \quad (0 < a < b), \quad (22)$$

where a, b denote the weights of a single additional product, and C is any fixed base assortment. We prove this for arbitrary C and T , not by induction on a collection of small horizons.

C.1 A coefficient lemma for truncated binomial rows

A sequence indexed by the integers is symmetric unimodal about a possibly half-integer center c if it is symmetric about c and nondecreasing to that center. Convolution preserves this property: decompose each sequence into positive combinations of centered interval indicators; the convolution of two interval indicators is symmetric and trapezoidal. Induction gives the assertion for any finite number of factors.

Lemma C.2 (A lower-half monotonicity fact). *Let $b_1, \dots, b_s$ be nonnegative integers. For each ℓ , choose an interval $[L_\ell, U_\ell] \subseteq [0, b_\ell]$ of integers with $L_\ell + U_\ell \geq b_\ell$, and define*

$$q_\ell(r) = \binom{b_\ell}{r} \mathbf{1}\{L_\ell \leq r \leq U_\ell\}.$$

*Then the convolution $Q = q_1 * \dots * q_s$, extended by zero, is nondecreasing on integer indices at or below $\frac{1}{2} \sum_\ell b_\ell$. Empty factors, and the empty convolution $Q = \mathbf{1}_{\{0\}}$, have the same property.*

Proof. Every nonempty superlevel set of q_ℓ is the intersection of $[L_\ell, U_\ell]$ with an interval symmetric about $b_\ell/2$. Its endpoints sum to at least b_ℓ . By the finite layer-cake decomposition, q_ℓ is a nonnegative combination of interval indicators centered at or to the right of $b_\ell/2$. Expand the convolution of these combinations. Each term is symmetric unimodal, with center at least $\frac{1}{2} \sum b_\ell$, and is therefore nondecreasing up to that location. Their nonnegative sum has the claimed monotonicity. A zero factor gives the zero sequence. $\square$

C.2 Multinomial weak-leader odds

Fix $m \geq 1$, an outside coordinate 0, and $r \geq 2$ inside coordinates. For positive weights $z = (z_0, \dots, z_r)$, define the homogeneous polynomial

$$A_i(z) = \sum_{\substack{u \in \mathbb{Z}_{\geq 0}^{r+1} \\ |u|=m}} \frac{m!}{\prod_h u_h!} z^u \mathbf{1}\{u_i \geq u_h \text{ for every } h = 1, \dots, r\}. \quad (23)$$

Normalized by $(\sum_h z_h)^m$, it is the probability that i is tied for largest among inside counts after m trials. Different leader events may overlap. This is *not* a uniformly tie-broken winner probability. In particular, all inside products are weak leaders when all observations are outside.

Theorem C.3 (Coefficientwise odds monotonicity). *For distinct inside indices i, j , every coefficient of*

$$(\partial_j A_i) A_j - A_i (\partial_j A_j) \quad (24)$$

is nonpositive. The coefficient of z_0^{2m-1} is $-m$. Hence, with all other weights fixed and $z_0 > 0$,

$$\frac{\partial}{\partial z_j} \frac{A_i(z)}{A_j(z)} < 0 \quad (z_j \geq 0).$$

Proof. By relabeling take $i = 1, j = 2$. Fix a combined count vector $t \in \mathbb{Z}_{\geq 0}^{r+1}$ with $|t| = 2m$ and $t_2 \geq 1$. Apart from the positive factor $(m!)^2 / \prod_h t_h!$, the coefficient of z^{t-e_2} in (24) is

$$\sum_{\substack{u \leq t, |u|=m \\ u_1 \geq u_h \ (h \geq 1) \\ t_2 - u_2 \geq t_h - u_h \ (h \geq 1)}} (2u_2 - t_2) \prod_h \binom{t_h}{u_h}. \quad (25)$$

This is a two-sample split: the first sample has leader 1, the second has leader 2. Let $B_t(k)$ be the sum of the same nonnegative binomial weights restricted to $u_2 = k$. It suffices to show

$$B_t(k) \leq B_t(t_2 - k) \quad (k > t_2/2). \quad (26)$$

Indeed, pairing k with $t_2 - k$ makes each contribution to (25) nonpositive.

Put $s = t_2 - k < k$ and fix also $u_1 = a$, $t_1 - u_1 = b$. In the slice with $u_2 = k$, feasibility requires

$$a \geq k, \quad b \leq s. \quad (27)$$

For another inside coordinate $h \geq 3$, the remaining constraints are

$$\max\{0, t_h - s\} \leq u_h \leq \min\{t_h, a\}.$$

The outside coordinate is unrestricted. Thus, after removing the factors $\binom{t_1}{a}\binom{t_2}{k}$, this slice has weight $Q_{a,s}(R)$, where

$$R = m - a - k,$$

and $Q_{a,s}$ is the convolution of the full outside binomial row and these truncated inside binomial rows. Since $a \geq k > s$, every nonempty truncation has endpoint sum at least t_h . Lemma C.2 applies with total

$$B = \sum_{h \neq 1,2} t_h = 2m - a - b - k - s.$$

Now exchange the two counts of coordinate 2, leaving a, b fixed. The first sample needs

$$R' = m - a - s = R + k - s$$

other observations. Each inside interval becomes $[\max\{0, t_h - k\}, \min\{t_h, a\}]$, a superset of its old interval. Thus the new convolution $Q_{a,k}$ dominates $Q_{a,s}$ coefficientwise. Furthermore,

$$2R' - B = b - a + k - s \leq 0$$

by (27). Both $R \leq R'$ lie below the midpoint, so

$$Q_{a,k}(R') \geq Q_{a,s}(R') \geq Q_{a,s}(R).$$

The unchanged factors match because $\binom{t_2}{k} = \binom{t_2}{s}$. Every old feasible a remains feasible after the exchange; the new slice may admit additional values of a . Summing proves (26).

This proves coefficientwise nonpositivity. For the strict coefficient, set $t_0 = 2m - 1, t_2 = 1$, with all other totals zero. The only admissible split has $u_2 = 0$: the first sample must be entirely outside to let product 1 be a weak leader. The second has one occurrence of product 2. Its contribution is exactly $-m$. Finally $A_j \geq z_0^m > 0$, so division by A_j^2 preserves strict negativity, including at $z_j = 0$. $\square$

C.3 An all-horizon exchange theorem

Fix a base assortment C with positive weights v_i , outside weight one, and set

$$D = 1 + \sum_{i \in C} v_i, \quad Z = D + x.$$

Let $f_C(x)$ denote its expected maximum load after adding a product of weight $x \geq 0$. Adding weight zero is the same as omitting the product. For $T \geq 2$, use the polynomials (23) with $m = T - 1$, the old weights fixed, and the new weight x . Denote the new product by $*$.

Lemma C.4 (Exact derivative).

$$f'_C(x) = \frac{T}{Z^{T+1}} \left(D A_*(x) - \sum_{i \in C} v_i A_i(x) \right) = \frac{T A_*(x)}{Z^{T+1}} \left(D - \sum_{i \in C} v_i \frac{A_i(x)}{A_*(x)} \right). \quad (28)$$

Proof. Differentiate the product distribution of the T customer outcomes, one customer at a time. Conditional on the other $T - 1$ outcomes, an additional inside choice of i increases the largest inside count by one exactly when i is a weak leader; an outside choice does not increase it. The probability derivatives are D/Z^2 for the newcomer and $-v_i/Z^2$ for an old product. The common baseline maximum cancels since all probability derivatives sum to zero. There are T possible differentiated customers, and the leader probabilities are A_i/Z^{T-1} . This gives (28). $\square$

Theorem C.5 (Strict endpoint exchange). *For every base assortment and every $T \geq 1$, f_C is strictly increasing or strictly decreasing up to one minimum and strictly increasing thereafter. In particular (22) holds for every $0 < a < b$.*

Proof. If C is empty, $f_C(x) = Tx/(1+x)$ is strictly increasing. If $T = 1$, then $f_C(x) = 1 - 1/(D+x)$, with the same conclusion. Otherwise, the bracket

$$B_C(x) = D - \sum_{i \in C} v_i A_i(x)/A_*(x)$$

in (28) is strictly increasing by Theorem C.3. Moreover, A_* has leading term x^{T-1} , whereas A_i has degree at most $\lfloor (T-1)/2 \rfloor$ in x , since i cannot be a weak leader with more newcomer occurrences than its own. Thus $B_C(x) \rightarrow D > 0$. The derivative has either no positive zero or a unique one, crossing from negative to positive. Its prefactor is everywhere positive. This proves the shape and the strict endpoint property, including the case $B_C(0) = 0$. $\square$

Proof of the structural part of Theorem C.1. Suppose an optimal assortment contains weight a but excludes weight $b > a$. Delete the former to obtain the base C . By Theorem C.5, either leaving it deleted or replacing it with weight b strictly improves the objective, a contradiction. Therefore every optimum has no strict inversion. Symmetry permits arbitrary relabeling within an equal-weight block, so one of the sorted prefixes is optimal. $\square$

Convexity is unnecessary and is in fact false. For base weights $(2, 4)$ and $T = 8$, let g_k be the conditional expected terminal maximum given exactly k purchases of the added product. Write $q = x/(7+x)$ and $f_C(x) = \sum_{k=0}^8 \binom{8}{k} q^k (1-q)^{8-k} g_k$. Exact values give

$$g_2 - 2g_1 + g_0 = -\frac{38600}{5764801} < 0.$$

Thus the q -polynomial has negative second derivative near zero. The proof uses monotonicity of a *scaled derivative*, not a nonnegative Hessian. It also does not require the sequence $(g_{k+1} - g_k)$ itself to be monotone.

C.4 Exact polynomial-time implementation

Scale all rational weights by a common denominator, giving positive integers $a_0, a_1, \dots, a_n$ in descending inside-weight order. Scaling does not change choice probabilities. For a cap $h \in \{0, \dots, T-1\}$, let $B_{k,h}(t)$ be the total integer weight of length- t sequences from the outside and first k products whose inside counts all remain at most h . Then

$$B_{0,h}(t) = a_0^t, \quad B_{k,h}(t) = \sum_{r=0}^{\min\{h,t\}} \binom{t}{r} a_k^r B_{k-1,h}(t-r). \quad (29)$$

The recurrence chooses the positions assigned to product k . It leaves the outside count uncapped. If $D_k = a_0 + \dots + a_k$, the exact prefix value is

$$F_T(\{1, \dots, k\}) = T - \frac{\sum_{h=0}^{T-1} B_{k,h}(T)}{D_k^T}. \quad (30)$$

Compute (29) for every h , retaining the value for every k , and compare the n rational numbers (30). There are $O(nT^3)$ arithmetic operations. The recurrence itself is a standard truncated generating-function calculation; the structural reduction to prefixes is the substantive optimization claim.

If the input bit length is $\mathcal{L}$, a common denominator has polynomial bit length. Each coefficient obeys $0 \leq B_{k,h}(t) \leq D_k^t$; hence it has $O(T(\mathcal{L} + \log n))$ bits. Powers, binomial coefficients, accumulated numerators, and cross-multiplied comparisons also have polynomial bit length. This proves the algorithmic part of Theorem C.1. The result is not asserted to be polynomial in the binary length of an exponentially large T .

For example, with five unit-weight products and $T = 2$, the prefix values are

$$1, \quad 10/9, \quad 9/8, \quad 28/25, \quad 10/9.$$

The optimum has three products: prefix optimality does not mean that one should always offer a singleton or the entire universe.

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